



What Makes a Level Hard?

We tried four ways to make behavior predict difficulty. Counting the pieces beat them all.

Alperen Akyol (alperen.akyol@std.bogazici.edu.tr) · **Advisor:** Bahri Atay Özgövde



Project Repo

The Problem Statement

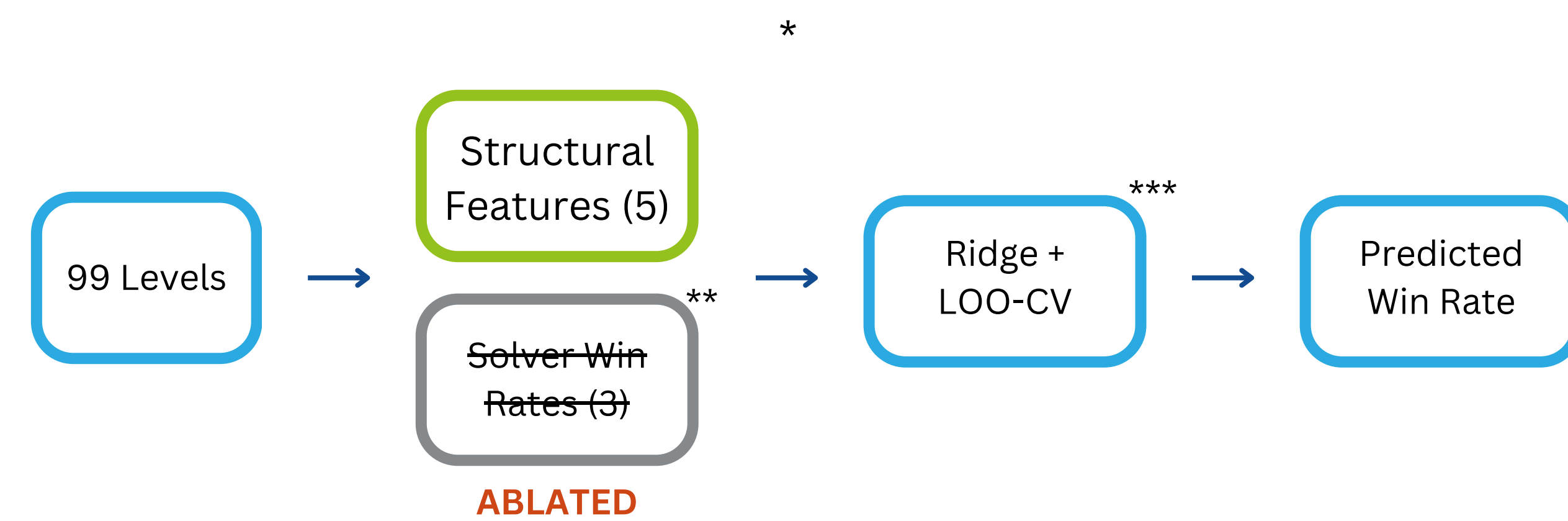
Predict a level's human win rate from the level alone — no pre-launch playtesting.

Why it matters: Procedural generation needs difficulty estimates before players see the level

Input: 99 real Hex Fall levels

Target: Real human win rate per level (from Paxie)

Bounded-Rationality Solver Ensemble



***Structural:** colours · total pieces · pins · ice · board size

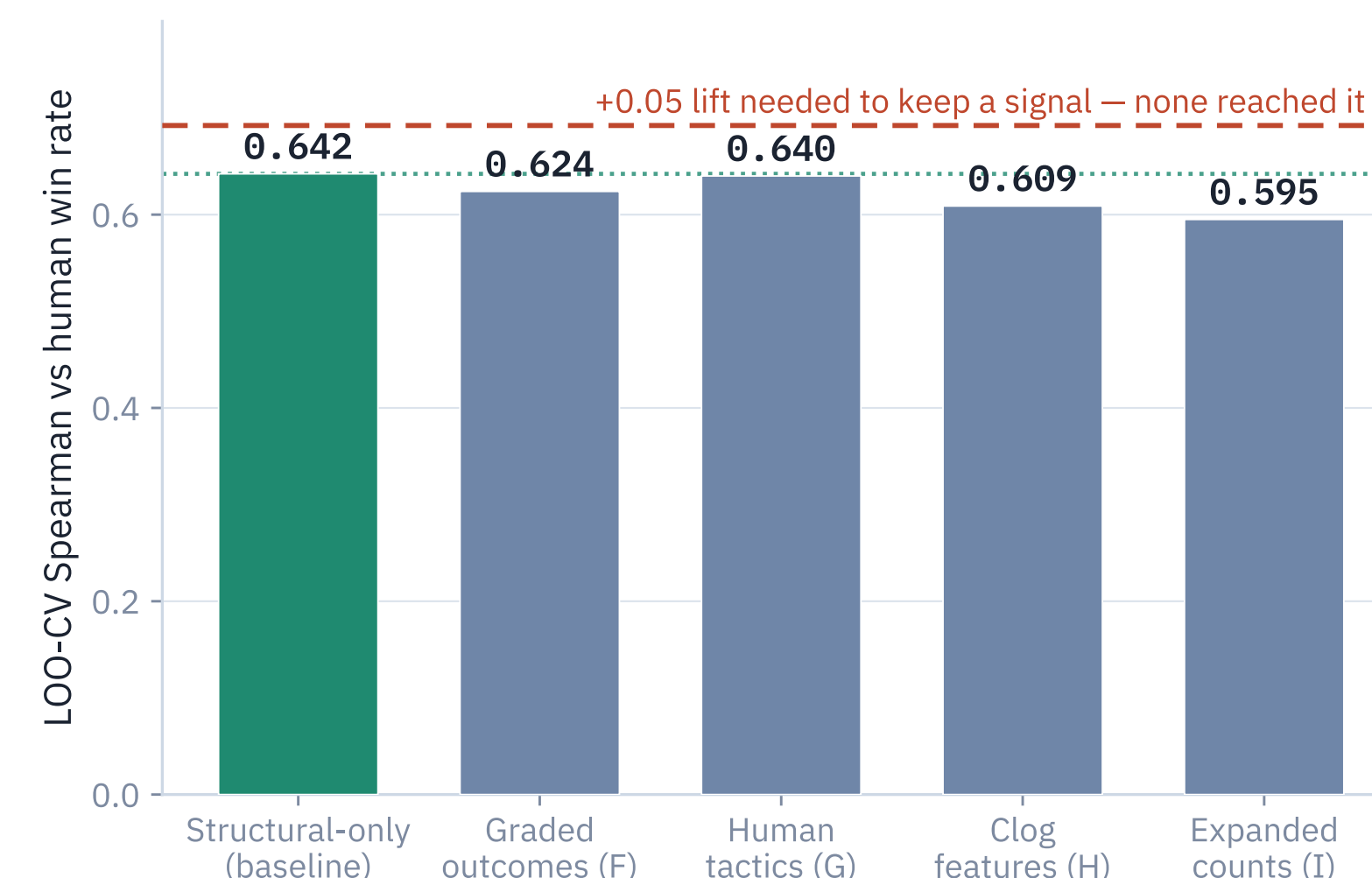
****Solvers:** greedy · lookahead · MCTS

*****Fit:** Ridge regression, leave-one-out cross-validation against 99 real human winrates

Ablation Across Four Signal Axes

Four pre-registered ablations tested whether any behavioural or decision-structure signal would improve on the structural-only baseline. The acceptance threshold was a Spearman lift of +0.05.

- Graded outcomes (F):** solver partial-progress metrics (fraction cleared, moves survived).
- Human-tactics player (G):** six-heuristic score approximating human play.
- Clog features (H):** Unified Clog Hypothesis metrics (forced-choice count, colour-transition density, hidden-cell ratio).
- Expanded structural counts (I):** additional structural measures (generators, walls, boxes, stack heights, mechanic diversity).



None reached the threshold. The Clog features were verified orthogonal to level size and yet contributed no predictive signal.

Initial Methodology and Validity Objection

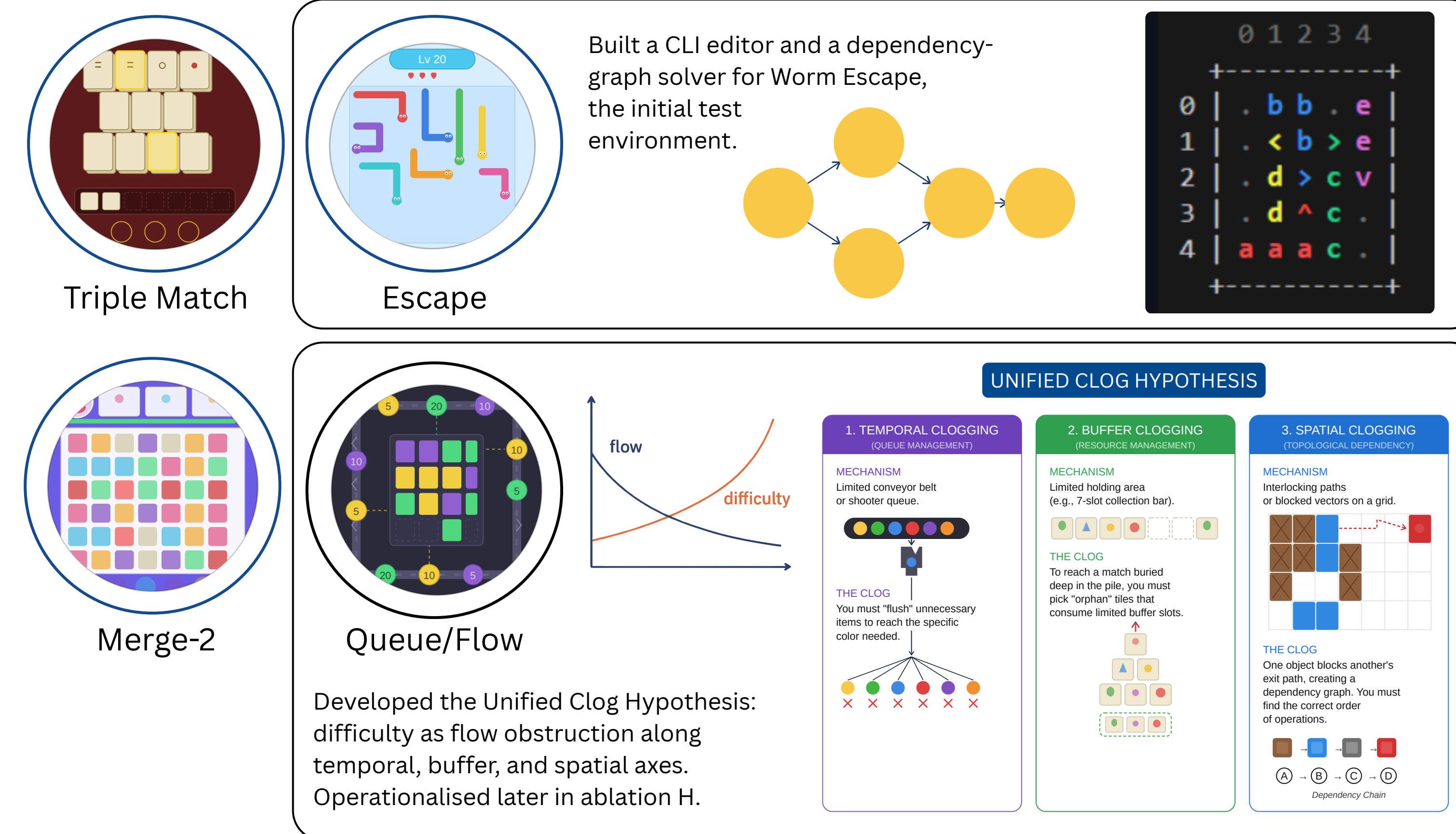
The original plan used a curiosity-driven PPO agent, with win rate as the difficulty signal. Before training began, the advisor raised three concerns:

- RL is disproportionate for a sparse-reward puzzle.
- RL agents converge toward optimal play, whereas humans do not, creating a validity mismatch between agent and human win rates.
- Reward shaping changes the objective being optimised rather than simulating different human skill levels.

Given these validity concerns, the agent-based approach was abandoned, although the simulation framework was retained.

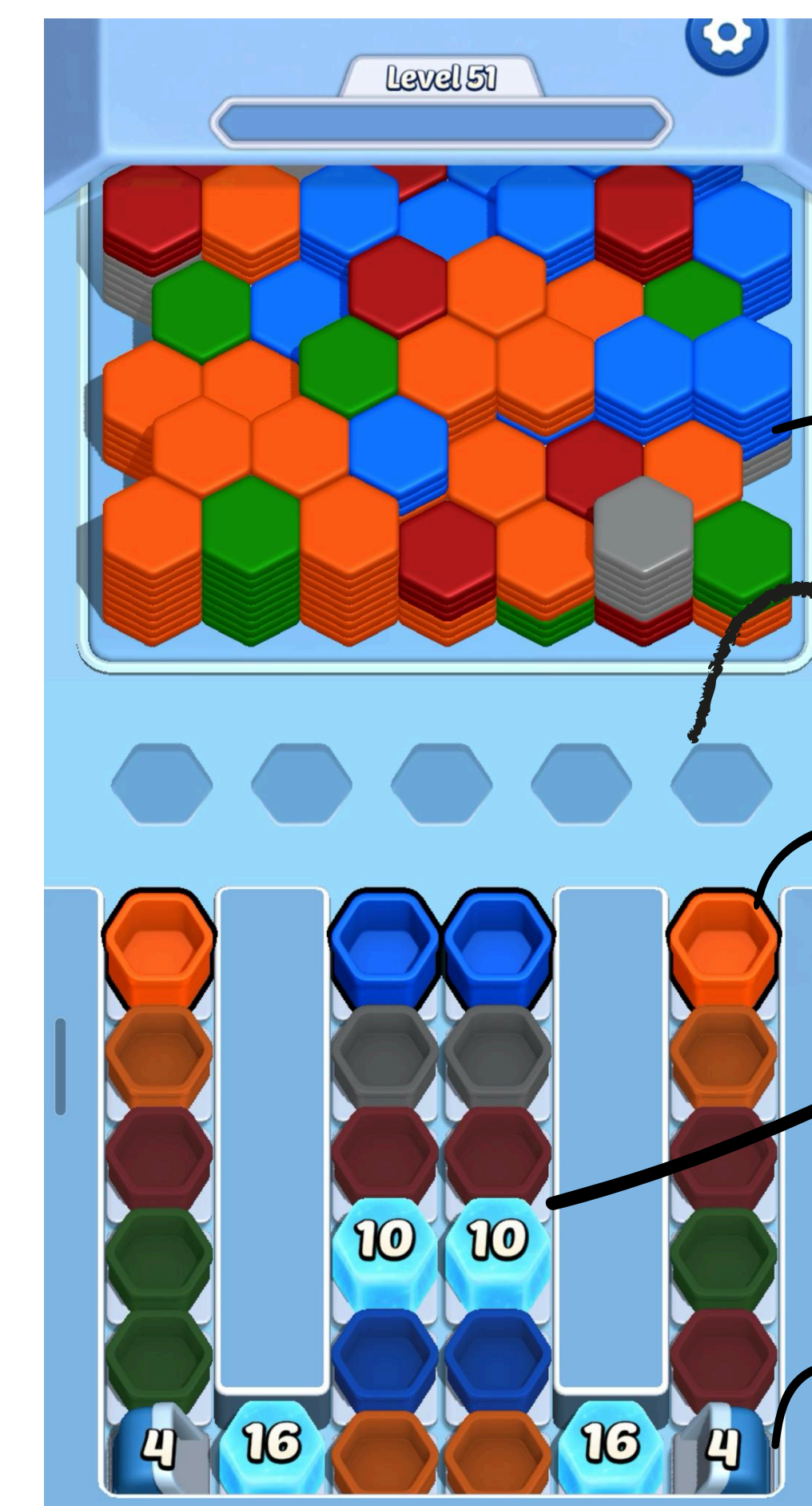
Early Phase

The project's first two months focused on a broader question: where in Paxie's portfolio is difficulty most tractable, and what makes a level hard in the first place?



The scope was narrowed to Hex Fall in early April due to higher corpus quality and the availability of per-level win-rate data provided by the Paxie CTO, which made it suitable for reliable measurement and comparative analysis.

Experimental Domain - Hex Fall



What is Hex Fall?

Hex Fall is a triple-match mobile puzzle game. The player sorts coloured slices into matching buckets while contending with several obstructions, before the board reaches saturation.

Slices: coloured pieces to be cleared.

Buffer: 5 slots for active buckets; filling all of them causes a lose state.

Bucket: Collects one colour up to 24 slices.

Ice Buckets: Frozen, thaws after a fixed number of moves.

Generator: Generates buckets in its facing direction.

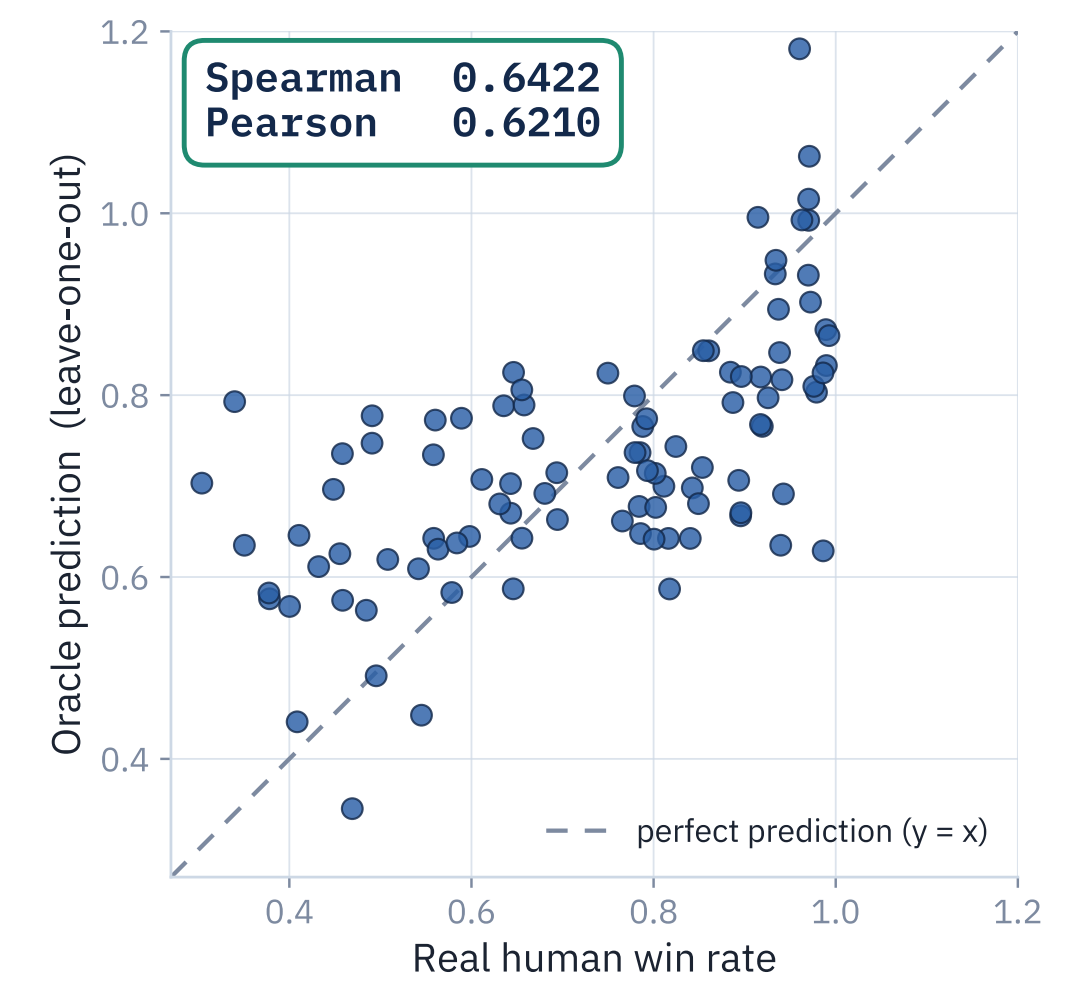
References & Acknowledgements

- Hüseyin Anil Özmen & Paxie Games (Hex Fall corpus and per-level win-rate data)
- Bergdahl, J., Gordillo, C., Tollmar, K., & Gisslén, L. (2020, August). *Augmenting automated game testing with deep reinforcement learning*.
- Ferdous, Raihana, et al. "Curiosity Driven Multi-agent Reinforcement Learning for 3D Game Testing."
- Hoerl, A. E., & Kennard, R. W. (1970). *Ridge regression: Biased estimation for nonorthogonal problems*.
- Kocsis, L., & Szepesvári, C. (2006, September). *Bandit based monte-carlo planning*.

Empirical Results

Five structural features predict human win rate on the 99-level corpus.

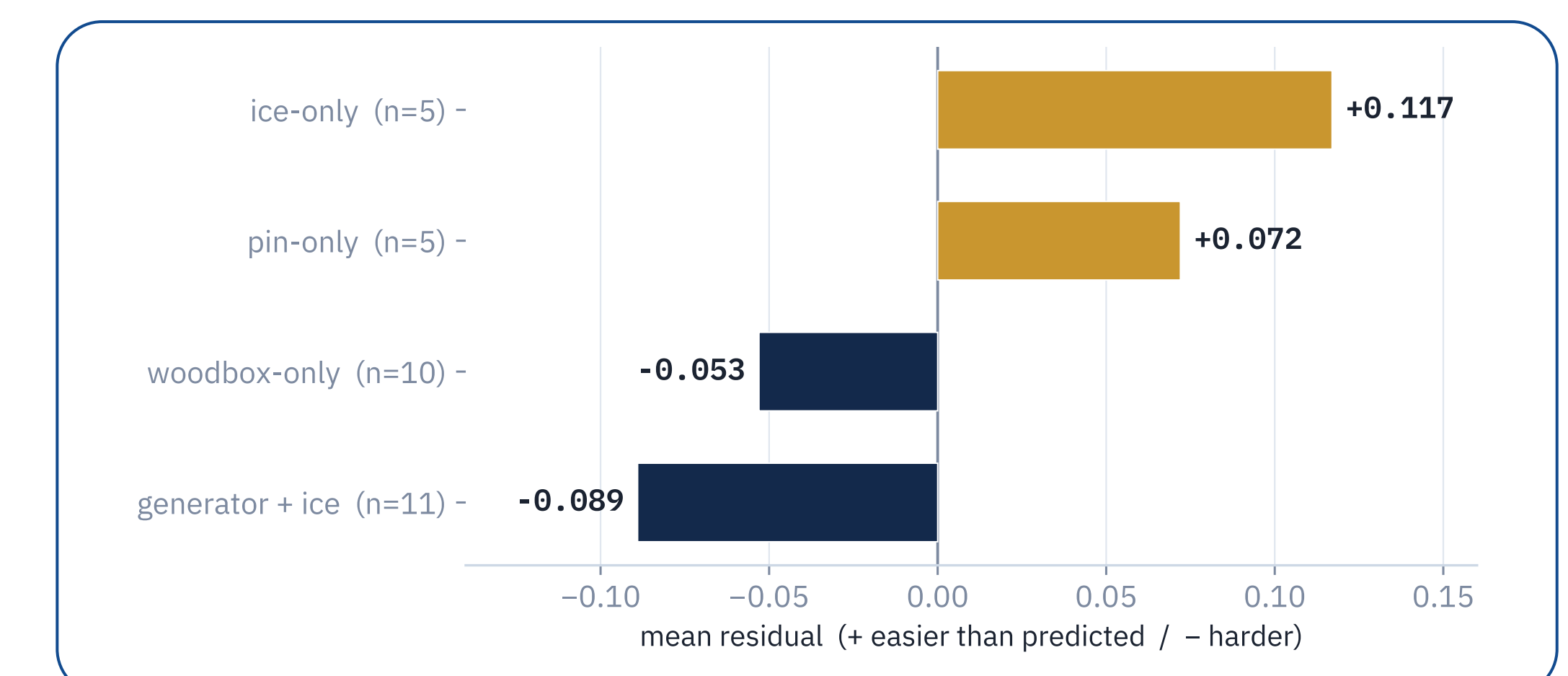
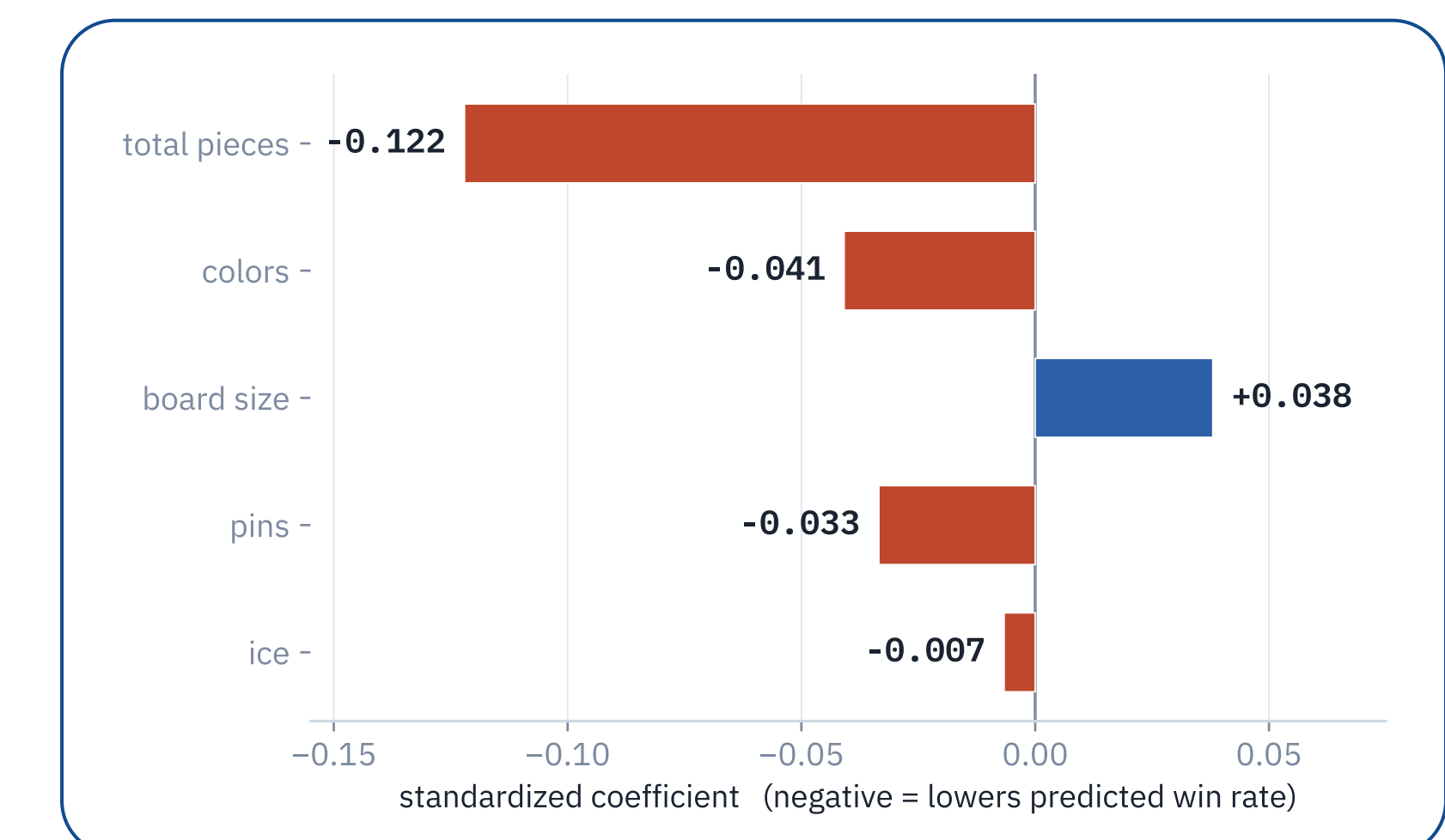
- LOO-CV Spearman 0.6422, Pearson 0.6210, on per-level win rate.
- No behavioural signal, no solver output, no tactical heuristic.
- Predicted versus real win rate, 99 levels (scatter right).



Feature Analysis

Predictive power is concentrated in a single feature, with residuals revealing mechanic-specific blind spots.

- Total slice count carries roughly three times the weight of any other coefficient (top).
- Even features verified orthogonal to slice count contribute no signal in regression.
- Difficulty on this corpus is dominated by board size.
- Residuals are not uniform across mechanics (bottom). Ice-only and pin-only levels are systematically easier than predicted; generator-plus-ice combinations are harder than the sum of parts.
- The structural features capture overall difficulty but miss specific mechanic interactions.



Limitations and Future Work

- The result is established on a single game and a single 99-level corpus. Generalisation across titles and corpora is not tested.
- Win rate saturates near 100 percent on easy levels. Average win duration is smoother and predicts at Spearman 0.96 on the same features.
- An open question is whether the structure-dominance finding holds in puzzle genres with richer decision structure, or is specific to Hex Fall.
- A second open question is whether an agent shaped toward characteristically human errors, rather than toward bounded optimality, could recover the predictive signal that bounded-rationality solvers did not.